Integrated motor drivers, dual buck supplies, encoder conditioning and Pico H.
EV3 jack land pattern is PROVISIONAL. Manufacturing release is blocked.
Use electronics/README.md and generated/electronics/release-status.json.

controller

File: controller.kicad_sch

power

File: power.kicad_sch

buck1

File: buck1.kicad_sch

buck2

File: buck2.kicad_sch

arm_R

File: arm_R.kicad_sch

arm_L

File: arm_L.kicad_sch

arm_F

File: arm_F.kicad_sch

arm_B

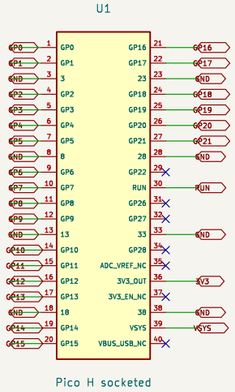
File: arm_B.kicad_sch

gonewx/quadcuber

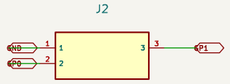
Sheet: /
File: quadcuber.kicad_sch

Title: Quadcuber four-arm integrated controller — A0 ENGINEERING HOLD

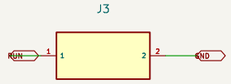
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 9.0.2		Id: 1/9



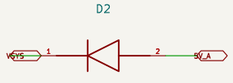
Pico H socketed



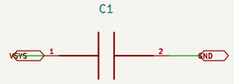
UART GND TX RX



RESET



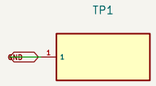
SS14 VSYS isolation



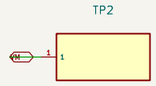
10u/25V



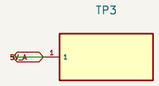
0.5A encoder PTC



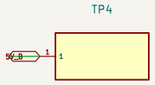
GND



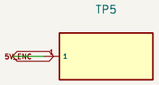
VM



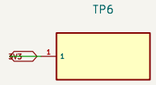
5V_A



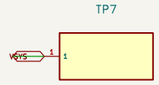
5V_B



5V_ENC



3V3



VSYS

Sheet: /controller/
File: controller.kicad_sch

Title: Quadcuber A0 / controller / ENGINEERING HOLD

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 9.0.2		Id: 2/9

J1



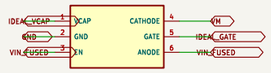
9V regulated input

F1



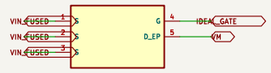
8A 32V fuse

U4



LM74700QDBVRQ1

Q1



CSD18540Q5B

C4



100n

C5



100n

D4



SMBJ12CA input transient

D3



SMBJ12A transient clamp

C2



470u/25V

C3



470u/25V

#FLG1



PWR_FLAG

#FLG2



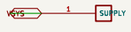
PWR_FLAG

#FLG3



PWR_FLAG

#FLG4



PWR_FLAG

#FLG5



PWR_FLAG

#FLG6



PWR_FLAG

#FLG7



PWR_FLAG

#FLG8

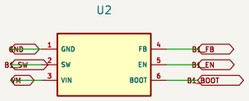


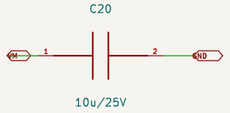
PWR_FLAG

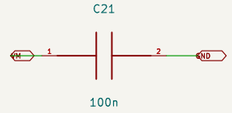
Sheet: /power/
File: power.kicad_sch

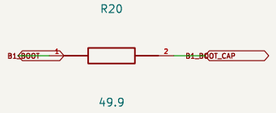
Title: Quadcuber A0 / power / ENGINEERING HOLD

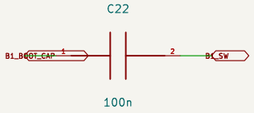
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 9.0.2		Id: 3/9

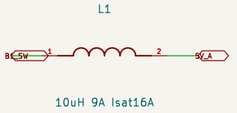


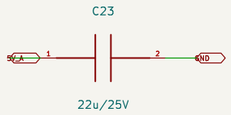


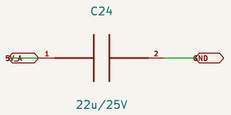


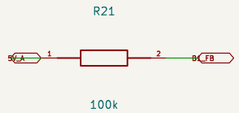


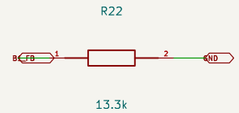


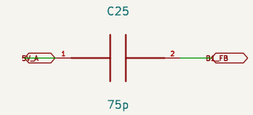


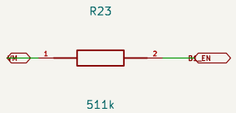


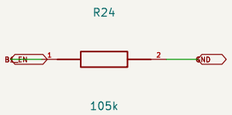


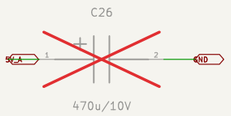








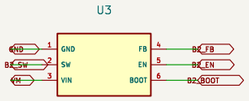


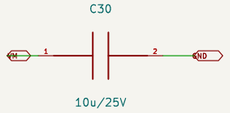


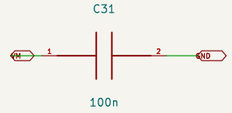
Sheet: /buck1/
File: buck1.kicad_sch

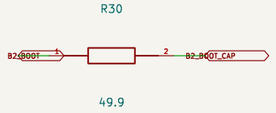
Title: Quadcuber A0 / buck1 / ENGINEERING HOLD

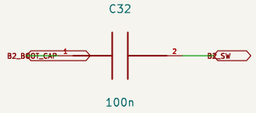
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 9.0.2		Id: 4/9

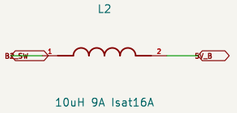


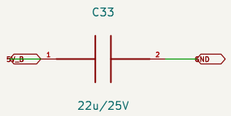


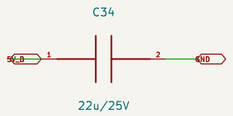


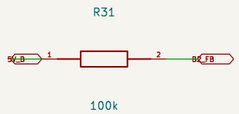


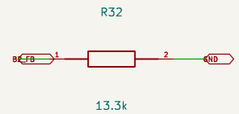


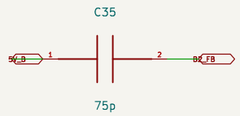


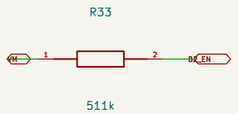


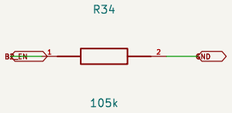


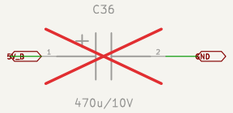










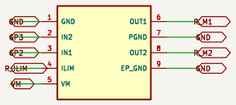


Sheet: /buck2/
File: buck2.kicad_sch

Title: Quadcuber A0 / buck2 / ENGINEERING HOLD

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 9.0.2		Id: 5/9

U10



DRV8871DDAR

R100



43.2k

R101



100k

R102



100k

C100



100n

C101



10u/25V

C102



470u/25V

J10



EV3 R 4434 HOLD

U20



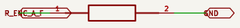
SN74LVC2G17DBVR

R103



1k

R105



100k

R104



1k

R106



100k

C103



100n

C104



100n

F10



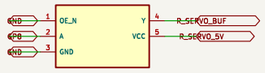
1.1A servo PTC

J20



R SERVO G + S

U30



SN74AHCT1G125DBVR

C105



100n

R109



100k

R107



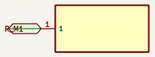
220

R108



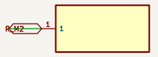
100k

TP10



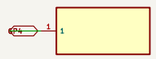
R_M1

TP11



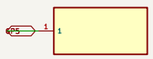
R_M2

TP12



GP4

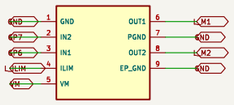
TP13



GP5

Sheet: /arm_R/	
File: arm_R.kicad_sch	
Title: Quadcuber A0 / arm_R / ENGINEERING HOLD	
Size: A3	Date:
KiCad E.D.A. 9.0.2	**Rev: A0**
	Id: 6/9

U11



DRV8871DDAR

R120



43.2k

R121



100k

R122



100k

C120



100n

C121



10u/25V

C122



470u/25V

J11



EV3 L 4434 HOLD

U21



SN74LVC2G17DBVR

R123



1k

R125



100k

R124



1k

R126



100k

C123



100n

C124



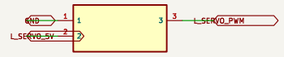
100n

F11



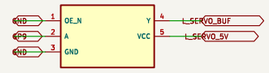
1.1A servo PTC

J21



L SERVO G + S

U31



SN74AHCT1G125DBVR

C125



100n

R129



100k

R127



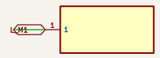
220

R128



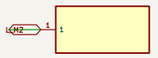
100k

TP14



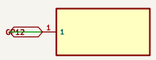
L_M1

TP15



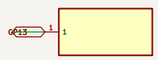
L_M2

TP16



GP12

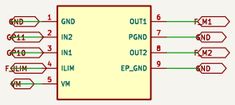
TP17



GP13

Sheet: /arm_L/		
File: arm_L.kicad_sch		
Title: Quadcuber A0 / arm_L / ENGINEERING HOLD		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 9.0.2		Id: 7/9

U12



DRV8871DDAR

R140



43.2k

R141



100k

R142



100k

C140



100n

C141



10u/25V

C142



470u/25V

J12



EV3 F 4434 HOLD

U22



SN74LVC2G17DBVR

R143



1k

R145



100k

R144



1k

R146

100k

C143

100n

C144

100n

F12

1.1A servo PTC

J22

F SERVO G + S

U32

SN74AHCT1G125DBVR

C145



100n

R149



100k

R147



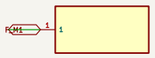
220

R148



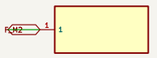
100k

TP18



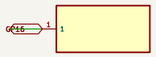
F_M1

TP19



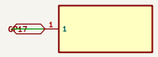
F_M2

TP20



GP16

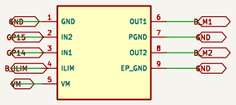
TP21



GP17

Sheet: /arm_F/		
File: arm_F.kicad_sch		
Title: Quadcuber A0 / arm_F / ENGINEERING HOLD		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 9.0.2		Id: 8/9

U13



DRV8871DDAR

R160



43.2k

R161



100k

R162



100k

C160



100n

C161



10u/25V

C162



470u/25V

J13



EV3 B 4434 HOLD

U23



SN74LVC2G17DBVR

R163



1k

R165



100k

R164



1k

R166



100k

C163



100n

C164



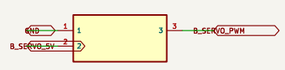
100n

F13



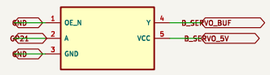
1.1A servo PTC

J23



B SERVO G + S

U33



SN74AHCT1G125DBVR

C165



100n

R169



100k

R167



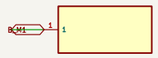
220

R168



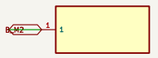
100k

TP22



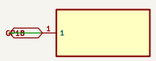
B_M1

TP23



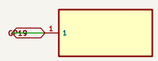
B_M2

TP24



GP18

TP25



GP19

Sheet: /arm_B/
File: arm_B.kicad_sch

Title: Quadcuber A0 / arm_B / ENGINEERING HOLD

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 9.0.2		Id: 9/9